

Driverless program budget

Complete bill of materials and resources to stand up the driverless program. All figures CAD (converted from USD at 1.37) - verify with live quotes.

BY CATEGORY

CATEGORY	COST (CAD)	LINE ITEMS
Compute	\$5,755	6
Safety / Rules Hardware	\$5,198	11
Test Equipment	\$3,252	7
Test Platform (RC)	\$2,959	1
Actuation - Brake / EBS	\$2,671	3
Networking / Power	\$2,123	4
Actuation - Steering	\$2,055	2
Simulation / HIL	\$1,644	2
Software / License	\$1,027	11
Facilities / Admin	\$685	1
Data / Storage	\$659	2
Sensing - Other	\$206	1
Sensing - LiDAR	\$0	1
Sensing - Camera	\$0	2
Sensing - GNSS/INS	\$0	2
Total	\$28,234	56

BY PRIORITY

PRIORITY	COST (CAD)	LINE ITEMS
Critical	\$18,233	30
High	\$9,111	20
Low	\$548	2
Medium	\$342	4
Total	\$28,234	56

Critical items are what the car needs to pass driverless technical inspection. LiDAR, cameras and GNSS/INS sit at \$0 because they are covered by the sponsorship asks on the last page.

BILL OF MATERIALS

ID	CATEGORY	ITEM / PURPOSE	RECOMMENDED MODEL / SPEC	QTY	UNIT COST (CAD)	EXT. COST (CAD)	PRIORITY	LEAD TIME	SPONSORABLE?	NOTES / RATIONALE
DV-001	Safety / Rules Hardware	Remote Emergency System (RES) - European events	Gross-Funk GF2000i-codec + T53R98, 2022 hw rev with E-Key	1	\$2,740	\$2,740	Critical	6-10 weeks	Possible (Ferchau sponsored Racetech's)	ORDER FIRST - longest lead item. Contact christian.keller@grossfunk.de / fsg.one/res. SIL3, CANopen Node-ID 0x011, 430-440 MHz.
DV-002	Safety / Rules Hardware	Remote Stop System - FSAE North America	FSAE Remote Stop: Vehicle Module + Control Box + Remote Stop Relay	1	\$685	\$685	Critical	4-8 weeks	No	NA uses its own standard, NOT the Gross-Funk unit. Design the shutdown circuit so either RES drops in.
DV-003	Safety / Rules Hardware	ASSI / DSSI status indicator lights	3x amber+blue LED panels, black background, >=15 cm2 shining area, LEDs <=20mm apart	3	\$116	\$348	Critical	2-3 weeks	No	States: Off=off, Ready=yellow steady, Driving=yellow flashing, Finished=blue steady, Emergency=blue flashing. Both sides behind cockpit + rear centreline.
DV-004	Safety / Rules Hardware	Driverless alert sound device	Automotive buzzer, 80-90 dBA, 1-5 Hz at 50% duty, 8-10 s	1	\$82	\$82	Critical	1-2 weeks	No	Sounds on Emergency state. Verify dBA at the rules-specified measurement distance.
DV-005	Safety / Rules Hardware	Autonomous Mission Indicator (AMI) display	Small dash display / LED set showing selected mission	1	\$110	\$110	Critical	1-2 weeks	No	Missions: acceleration, skidpad, trackdrive, braketest, inspection, autocross. Mount near dash/ASMS.
DV-006	Safety / Rules Hardware	ASMS / DSMS master switch	Direct-acting rotary switch, blue circular area >=50mm, labelled 'DS', lockout/tagout capable	1	\$164	\$164	Critical	2 weeks	No	Must be direct-acting (not via relay/logic). When OFF the car must remain normally drivable and pushable.
DV-007	Safety / Rules Hardware	DV shutdown-circuit relay set	Driverless System Relay (NO permissive) + Remote Stop Relay + NC bypass relay + EBS Relay (fail-open)	1	\$411	\$411	Critical	2-3 weeks	No	Automotive-rated relays; document the schematic for scrutineering.
DV-008	Safety / Rules Hardware	DV livery, signage and required markings	Vinyl decals, 'Brake Release' labels, DV class markings	1		\$0	High	1-2 weeks	Likely (local print shop)	Check the current-year rulebook for exact wording and placement.
DV-009	Software / License	Rulebook + standards library	ISO 26262 and ISO 21448 (SOTIF) - purchase or university library access	1	\$0	\$0	High	1 week	n/a	ISO copies are expensive (~US\$200-2,000/part). CHECK THE UNIVERSITY LIBRARY FIRST - usually free access.
DV-010	Software / License	Git hosting + CI minutes	GitHub Education (free private repos + Actions) or self-hosted GitLab	1	\$0	\$0	Critical	Immediate	Free for students	Apply via GitHub Education. Zero cost - decide the version-control strategy in month 1.
DV-011	Software / License	MATLAB / Simulink + toolboxes	Vehicle Dynamics Blockset, MPC Toolbox, Automated Driving Toolbox	1	\$0	\$0	High	2-4 weeks	Yes - MathWorks sponsors FS teams	MathWorks gives free competition licences to Formula Student teams. Apply early; also check your campus site licence.
DV-012	Software / License	CAD / PDM (already sponsored)	SOLIDWORKS / 3DEXPERIENCE via SAE Collegiate Design Series	1	\$0	\$0	Medium	Already held	Yes - Dassault via SAE	Apply under a TEAM email, not a personal one. Export data at end of season.
DV-013	Sensing - LiDAR	Primary LiDAR	Livox Mid-360 - 360x59 deg, 40m @10% / 70m @80%, 265g, IP67, built-in IMU, 0.1m blind zone	1		\$0	Critical	3-6 weeks	Possible via distributor	Best price/performance for a new team. Non-repetitive scan needs point accumulation. Weak vendor support.
DV-014	Sensing - Camera	Stereo camera	Used Stereolabs ZED 2i (rolling shutter, IP66, built-in IMU) or RealSense D455	1		\$0	Critical	1-3 weeks	Yes - Stereolabs has a student programme	Rolling shutter smears at speed - acceptable at the <5 m/s year-1 target. Verify RealSense availability; that line has had discontinuation uncertainty.
DV-015	Sensing - Camera	Lens set for machine-vision cameras	C-mount lenses sized for cone detection at 25m (far ~50-60 deg HFOV, near ~120 deg)	2		\$0	High	3-5 weeks	Possible	A 325mm cone at 25m on 1080p/50 deg HFOV is only ~15px tall - near the detection floor. Do the FOV maths before ordering.
DV-016	Test Equipment	Calibration targets	Rigid checkerboard board + printed AprilTag boards + calibration jig	1	\$274	\$274	High	2 weeks	No	Use with Kalibr (camera-IMU), lidar_camera_calibration, direct_visual_lidar_calibration. Rigidity matters more than size.
DV-017	Compute	ML training workstation	Ryzen 9 / i9, 64GB RAM, RTX 4080 (16-24GB), 2TB NVMe	1		\$0	High	2-4 weeks	Yes - NVIDIA academic seeding	Tier 1: use existing lab machines or campus HPC (cost \$0). Needed to train YOLO/segmentation on FSOCO.
DV-018	Software / License	Cone dataset	FSOCO - 44,195 images / 316,669 labelled cones, bboxes + instance segmentation	1	\$0	\$0	Critical	Immediate	Free	PUBLIC since Aug 2022 - the old contribution buy-in is gone. fsoco-dataset.com. Contributing back is still encouraged.
DV-019	Sensing - GNSS/INS	GNSS/INS	ArduSimple simpleRTK2B (u-blox ZED-F9P) + antenna, paired with a separate IMU	1		\$0	High	2-4 weeks	No	Budget RTK. No dual-antenna heading - yaw at standstill will be weak, which hurts skidpad starts.
DV-020	Sensing - GNSS/INS	Dual GNSS antennas + low-loss cabling	L1/L2 survey antennas x2 + LMR cabling, roll-hoop/bodywork mounts	1		\$0	High	3-4 weeks	Bundled with INS sponsorship	Antenna baseline length drives heading accuracy - plan the mounting geometry with the mech team.
DV-021	Software / License	RTK correction service (Canada)	Commercial NTRIP subscription (RTKdata / RTK FIX) - annual	1		\$0	Medium	1 week	Possible	Tier 1: free casters RTK2go / Emlid Caster (\$0). Only Quebec offers free provincial corrections - a Manitoba team will run its own base or pay. Cross-check NRCan's RTK network list.
DV-022	Test Equipment	Optical ground-speed sensor (ground truth)	Kistler Correvit (or GeneSys ADMA) - slip-free velocity reference	1		\$0	Low	10-16 weeks	Yes - usually sponsored	Validation only, shared with Testing. AMZ and Elefant both ran ground-speed sensors. Skip unless sponsored.
DV-023	Test Equipment	Survey-grade RTK rover for map ground truth	Handheld RTK rover to survey true cone positions on the test track	1		\$0	High	3-4 weeks	No	You cannot report a mapping RMSE without surveyed cone truth. Target <1m year 1; AMZ achieved 0.29m.

ID	CATEGORY	ITEM / PURPOSE	RECOMMENDED MODEL / SPEC	QTY	UNIT COST (CAD)	EXT. COST (CAD)	PRIORITY	LEAD TIME	SPONSORABLE?	NOTES / RATIONALE
DV-024	Software / License	FSAE Tire Test Consortium membership	TTC membership - measured Calspan tire data for Pacejka/Magic Formula fitting	1	\$685	\$685	High	2-4 weeks	No (team pays)	Essential for a credible vehicle model. Join at fsaettc.org. Cheapest single item with the biggest effect on model fidelity.
DV-025	Software / License	Optimisation and MPC solver stack	acados, CasADi, OSQP, HPIPM, TUM global_racetrajectory_optimization, GTSAM, Ceres	1	\$0	\$0	Critical	Immediate	Free / open source	All open source. TUM's minimum-curvature raceline optimiser is the standard offline tool; ETH's MPCC is the reference racing controller.
DV-026	Test Equipment	Rack-force load cell	Inline load cell + amplifier to measure real steering rack force before sizing the actuator	1	\$342	\$342	High	2-3 weeks	No	Size the steering actuator from MEASURED rack force, not motor datasheet headlines. Shared item with the Mech/EE interface work.
DV-027	Compute	Onboard compute	NVIDIA Jetson AGX Orin 64GB Developer Kit - 275 TOPS, 15-60W	1	\$1,369	\$1,369	Critical	3-8 weeks	Yes - NVIDIA seeds Jetsons to student teams	List US\$1,999 but frequently ~US\$999 on promotion. Minimises power/heat/package/pain vs x86. DisplayPort only.
DV-028	Data / Storage	Onboard logging NVMe	2TB NVMe SSD for rosbag/MCAP capture	1	\$247	\$247	Critical	1-2 weeks	No	LIDAR + multi-camera logs run several GB/min. Expect 1-2TB per full test day - offload nightly.
DV-029	Networking / Power	CAN interface	PEAK PCAN-USB / SocketCAN adapter x2	2	\$171	\$342	Critical	1-2 weeks	No	You must emit the FSG CAN message set (0x500, 0x501, 0x502, 0x511-0x515) at 100ms - design the DBC around it from day one.
DV-030	Networking / Power	Managed automotive Ethernet switch + PoE	Industrial managed switch with PoE for sensors, PTP-capable	1	\$548	\$548	Critical	2-4 weeks	No	Route Ethernet away from HV cabling; shield and ground for tractive-system EMI.
DV-031	Networking / Power	Connector + crimp tooling kit	Deutsch DTM / Autosport / TE / Amphenol kit plus proper crimpers	1	\$822	\$822	Critical	2-3 weeks	Possible	Bad crimps cause more DNFs than bad algorithms. Buy the real crimp tools.
DV-032	Networking / Power	Shielded Ethernet / GMSL2 cabling + glands	Shielded cabling, right-angle connectors, strain relief	1	\$411	\$411	High	2-3 weeks	No	Keep sensor data lines physically separated from HV runs.
DV-033	Compute	Microcontroller dev boards	STM32 Nucleo / Teensy 4.1 x5 for real-time control and state machine work	5	\$55	\$275	High	1-2 weeks	No	Real-time control belongs on MCUs, not on the Jetson. Run PREEMPT_RT on the main compute.
DV-034	Compute	Safety-oriented MCU dev kit	NXP S32K automotive MCU dev kit for EBS supervisor logic	1	\$206	\$206	Critical	2-3 weeks	Possible - NXP	Pairs with the non-programmable logic latch; scrutineers will ask to see the failure analysis.
DV-035	Safety / Rules Hardware	EBS supervisor board (non-programmable logic)	74xx-series latch + plausibility monitoring, custom PCB fab + assembly	1	\$548	\$548	Critical	4-6 weeks	No	The FSG reference design (Stollberger/Gebhardt) specifies NON-PROGRAMMABLE logic for the EBS latch. Do not implement this purely in software.
DV-036	Safety / Rules Hardware	DV interface PCB (fab + assembly)	Custom board tying shutdown circuit, relays, ASSI, AMI, sensors (JLPCB/PCBWay, 3 spins)	1		\$0	Critical	3-5 weeks per spin	No	Budget three revisions. Order spares of every connector on it.
DV-037	Sensing - Other	Steering angle absolute encoder	Magnetic absolute encoder (RLS / Bourns / CUI) on column or rack	1	\$206	\$206	Critical	2-3 weeks	No	Rules require logging steering angle actual vs target. Absolute, not incremental.
DV-038	Compute	Compute cooling + vibration isolation	Ventilated enclosure, fans, heatsinks, wire-rope isolators	1	\$891	\$891	High	3-4 weeks	No	Elefant used ventilated side boxes. Keep the compute off sprung-mass hot spots.
DV-039	Actuation - Steering	Steering actuator	Salvaged automotive EPS column motor (Prius/Corolla/VW) + ODrive or custom driver	1	\$1,233	\$1,233	Critical	4-8 weeks + reverse engineering	No	Cheapest hardware, most software risk - you must reverse-engineer the CAN/torque-sensor protocol. Manual steering must remain possible with ASMS OFF and nothing removed.
DV-040	Actuation - Steering	Steering drive train hardware	Ball screw, HTD belt + pulleys, machined bracket, bearings, shaft stock	1	\$822	\$822	Critical	4-6 weeks	Possible (machining sponsor)	Included in the Tier 2/3 actuator packages above; separate line for the Tier 1 DIY route.
DV-041	Actuation - Brake / EBS	EBS pneumatic system	Air tank + regulator + normally-open solenoid + Festo/SMC cylinder + 3x pressure sensors + fittings	1	\$1,644	\$1,644	Critical	6-10 weeks	Possible - Festo/SMC	Must use PASSIVE mechanical energy storage and release without electrical power. >=10 m/s2 average deceleration, <=200ms reaction. NO push-in fittings at the EBS.
DV-042	Actuation - Brake / EBS	Driverless master cylinders + shuttle valves	Dedicated DV master cylinders + 3-way higher-pressure-wins valves into existing hydraulic lines	1	\$685	\$685	Critical	4-6 weeks	Possible	Pattern used by Carnegie Mellon Racing and Politecnico di Torino.
DV-043	Safety / Rules Hardware	Brake Release deactivation handles	Max 2 red handles marked 'Brake Release', operable with no power or tools	2	\$55	\$110	Critical	2 weeks	No	Must be near the vehicle centreline on top of the car.
DV-044	Actuation - Brake / EBS	Hydraulic lines, fittings and brake spares	Braided lines, unions, bleed kit, spare pads/fluid	1	\$342	\$342	High	2-3 weeks	No	You will bleed this system many, many times during EBS validation.
DV-045	Test Platform (RC)	F1TENTH platform	Traxxas Slash 4x4 + VESC 6 MkVI + RPLiDAR A2/S2 + Jetson Orin Nano + USB global-shutter cam + 2x LiPo + deck/IMU/power board	1	\$2,959	\$2,959	Critical	4-6 weeks	Partial - NVIDIA for the Orin Nano	DO THIS FIRST. Exercises the entire ROS2 stack indoors over the Winnipeg winter, before the car is ready. BOM documented at f1tenth.org.
DV-046	Test Equipment	Practice cones	200x generic 12.5in traffic cones in blue/yellow/orange + 20x 18in orange	220	\$7	\$1,540	Critical	2-3 weeks	Possible (local supplier)	Max cone spacing in the driving direction is 5m, so a full autocross/trackdrive layout needs 200-400+. Buy cheap locally for practice.
DV-047	Test Equipment	Telemetry link	RFD900 / Ubiquiti one-way telemetry radio pair + antennas	1	\$548	\$548	High	2-4 weeks	No	One-way telemetry only - two-way control is banned between start/finish lines.
DV-048	Simulation / HIL	CI/CD runner machine	Dedicated self-hosted runner for ROS2 build + regression suite	1	\$1,644	\$1,644	High	2-4 weeks	No	Tier 1: GitHub Actions free education minutes (\$0). Testing responsibility #3 on the org chart.

ID	CATEGORY	ITEM / PURPOSE	RECOMMENDED MODEL / SPEC	QTY	UNIT COST (CAD)	EXT. COST (CAD)	PRIORITY	LEAD TIME	SPONSORABLE?	NOTES / RATIONALE
DV-049	Facilities / Admin	Indoor winter test space	Gym / lab / hangar booking for RC-scale testing Nov-Apr	1	\$685	\$685	High	Book 3-6 months ahead	Often free via the university	Winnipeg gives you roughly a 5-month outdoor blackout. Book indoor space early - this is the difference between a productive and a wasted winter.
DV-050	Simulation / HIL	Simulation stack	FSDS (UE4+AirSim), FSSIM, EUFS Sim, CARLA, AWSIM, Gazebo	1	\$0	\$0	Critical	Immediate	Free / open source	FSDS is the FS-specific standard (used for FS-Online 2020). FSSIM predicted gotthard's FSG 2018 trackdrive lap time to 1%. LGSVL/SVL is deprecated - do not start there.
DV-051	Compute	Simulation / visualisation workstation	RTX 4070+ build, 32-64GB RAM, for UE4-class sim and data viz	1	\$3,014	\$3,014	High	2-4 weeks	Yes - NVIDIA academic	Tier 1: reuse existing lab PCs (\$0). FSDS is UE4-based and needs a real GPU.
DV-052	Data / Storage	Test data storage	2x 4TB external USB SSD/HDD for test-day offload	2	\$206	\$412	Critical	1-2 weeks	No	Storage architecture and data integrity are Sim responsibilities #2. Expect 1-2TB per full test day.
DV-053	Software / License	Visualisation and log tooling	Foxglove Studio, PlotJuggler, rerun.io, MCAP tooling	1	\$0	\$0	High	Immediate	Free tiers	Foxglove has a free tier; the paid team tier is optional. MCAP is the modern rosbag format.
DV-054	Software / License	Labelling platform	CVAT (self-hosted, free) or Roboflow paid tier	1	\$342	\$342	Medium	1 week	Possible	Sim responsibility #6 (multi-modal labelling). Self-hosted CVAT is \$0 if you have a spare machine.
DV-055	Software / License	Commercial simulation (sponsorship route)	IPG CarMaker, Ansys AVxcelerate, dSPACE ASM, MATLAB RoadRunner	1	\$0	\$0	Medium	8-12 weeks	YES - all four run student/FS programmes	RoadRunner is free for education. CarMaker/AVxcelerate/dSPACE are five-figure licences that FS teams routinely get sponsored - apply, do not budget cash.
DV-056	Test Equipment	Driver-in-the-loop input rig	Force-feedback wheel + pedals for human-baseline comparison runs	1	\$548	\$548	Low	2-3 weeks	No	Nice-to-have for benchmarking the autonomous line against a human reference.

SPONSORSHIP TARGETS

VENDOR	WHAT TO ASK FOR	OWNING ROLE	WHY THEY ARE A FIT	HOW TO APPROACH
SBG Systems	Ellipse-D INS/GNSS	Mapping	Runs an education/sponsorship programme; used by AMZ, Chalmers, TUfast, Zurich UAS. ITAR-free.	Apply 6-9 months ahead via their Formula Student form; lead with your project plan and timeline.
VectorNav	VN-300 dual-GNSS/INS	Mapping	Supports student teams; 0.15 deg static / 0.2 deg dynamic heading at 400Hz.	Alternative to SBG — apply to both, take whichever lands.
NVIDIA	Jetson AGX Orin, RTX GPUs	Firmware / Perception	Academic hardware seeding programme.	Apply through the NVIDIA academic/developer programme; a strong technical proposal matters.
MathWorks	MATLAB, Simulink, VD Blockset, MPC Toolbox	Team-Wide	Gives free competition licences to Formula Student teams.	Nearly automatic for registered FS teams — do this in month 1.
Speedgoat	Real-time target / HIL rig	Testing	Explicitly sponsors Formula Student; lists AMZ and Greenteam as references.	Ask for the HIL rig rather than budgeting US\$8k cash.
Ouster	OS1-32 / OS1-128 LiDAR	Perception	Non-profit pricing (~US\$6k vs US\$8k list); absorbed Velodyne.	Ask for both a discount and a loaner; used units on the secondary market are the fallback.
Stereolabs	ZED 2i / ZED X cameras	Perception	Runs a student programme.	Low-cost ask with a high hit rate — request the ZED X for global shutter.
Vector Informatik	CANoe / CANalyzer, VN interfaces	Firmware	University and FS programme.	Ask for the software licence and the hardware interface as a bundle.
IPG Automotive	CarMaker	Sim	Student programme for FS teams.	Five-figure licence normally — never budget cash for this.
dSPACE	HIL / ASM simulation	Testing / Sim	University programme.	Higher barrier; approach after you have a track record.
Ansys	AVxcelerate, simulation suite	Sim	Well-established student team programme.	Bundle the ask with your mechanical/aero subteam's needs.
Maxon	EC motors, planetary gearheads, EPOS4	Mech/EE Interface	Documents FS steering and rear-wheel-steer applications.	Bring measured rack-force data to the conversation — they size the motor with you.
Festo / SMC	Pneumatic cylinders, valves, regulators	Mech/EE Interface	Standard EBS component suppliers for FS teams.	Ask for the EBS kit as an in-kind donation; include your FMEA.
Dassault / SOLIDWORKS via SAE	CAD + 3DEXPERIENCE PDM	Team-Wide	SAE Collegiate Design Series sponsorship.	Keep it under a team email; export data at end of season.
Gross-Funk	RES (GF2000i-codec / T53R98)	DSO	Mandated unit; Ferchau has sponsored a team's RES before.	Order regardless of sponsorship outcome — the lead time is the binding constraint.
AiM / MoTeC	DAQ, lap timing	Testing	Both support FS teams.	You may already have a relationship through the EV car.
Local machine shops	CNC sensor mounts, brackets	Perception / Mech-EE	Winnipeg-area shops often support student teams in kind.	In-kind machining is the most under-used sponsorship channel.